

Objective and significance. Adaptive instructional systems (AIS) often adjust task difficulty with a learned policy that runs behind hand-written constraints. When the two are evaluated together, an outcome cannot be attributed to the learned component, and simpler alternatives are seldom reported under the same constraints. Simulation studies of adaptive mock-interview tutoring compare learned policies with heuristics [1], and reviews of reinforcement learning in education report inconsistent baselines and heavy reliance on simulated users [2]. In the literature we reviewed, a matched comparison against a state-blind constant action under an identical constraint layer, with an equivalence analysis and an account of how often the constraints act, is less often reported. We ask one research question: under an identical application-level rule-based guardrail, what does a learned proximal-policy-optimisation (PPO) difficulty controller add beyond a state-blind Constant-Same policy in simulated technical-interview sessions? The study is simulation-only. No learners or real candidates took part, and it makes no claim about learning effectiveness, learner outcomes or superiority of the learned policy. Its relevance to AIS is methodological: what evidence is needed before an adaptive decision can be credited to a learned component.

Methods and approach. A simulated candidate has a persona type and a skill, and produces performance, confidence, hesitation and response-time signals for the current difficulty. The controller observes a six-dimensional state and chooses among Easier, Same and Harder; difficulty lies between 1 and 5 and starts at 3.0. The primary stratum is 40 authored personas (five types by eight skill levels), each with an authored target difficulty; a session has ten turns. The endpoint is the persona-level difference in mean absolute error between session difficulty and target difficulty. Five frozen PPO checkpoints (five training seeds, each trained for 24,576 timesteps against one default simulated candidate) were compared with Constant-Same over 20 evaluation seeds, both under the same guardrail: six ordered rules that hold, ease or raise difficulty from score, confidence and hesitation thresholds, with no formal safety specification. The persona is the statistical unit and the training seed is a second random factor; the interval is a two-way cluster bootstrap (10,000 resamples). The equivalence margin of ± 0.12 difficulty levels and the superiority threshold of -0.20 were pre-specified in the project repository and author-selected; no external empirical justification was identified. Protocol and harness were committed and tagged before the run; the tags are local and no external registry entry was made. Three further pre-specified analyses (persona-only bootstrap, seed-level t interval, leave-one-seed-out) give seven sensitivity estimates. A separate evaluation-only replay of the frozen checkpoints on five fixed personas (125 PPO sessions, 1250 turns) accounts for the guardrail and for behavioural divergence.

Results and findings. The difference in tracking error (PPO minus Constant-Same) was -0.0350 (95% interval -0.0818 to $+0.0021$; mean errors 1.0688 and 1.1039), classified equivalent within the margin. Superiority was not met, and all seven sensitivity estimates were classified equivalent. Equivalence does not mean identical behaviour or no effect. In the replay, the guardrail activated on 563 of 1250 PPO turns (45.0%); 99 activations changed the selected action (7.9% of all turns) and 464 were no-op activations. At least one override occurred in 41 of 125 sessions. Relative to Constant-Same under the same guardrail, the final-action sequence differed in 50 of 125 sessions, the executed difficulty path in 25 and session tracking error in 5; these counts follow different definitions. On the 40-persona grid, guarded PPO was more volatile than Constant-Same (difference $+0.149$, interval 0.067 to 0.254). Replacing PPO's observation with zeros or another persona's observation raised tracking error, so the policy reads its state, but that does not show that reading it improves tracking.

Contributions and implications. The study contributes a matched-policy comparison in which the constraint layer is held fixed, an equivalence result with its pre-specified rule and margin, and a guardrail accounting that separates rule activations from action overrides and from attempted boundary actions. For evaluating adaptive instructional systems, the implication is to report constraint-layer accounting and a matched simpler baseline before attributing an outcome to a learned policy, and to read equivalence as a statement about its stated margin. Limitations bound every finding. The simulator, personas, target rule and margin are authored by the study team; the training reward contains an alignment term with an authored rule; the checkpoints were trained against one

simulated candidate under training settings that differ from evaluation, and the application's deployed policy and state definition differ from the evaluated ones; no Elo, item-response or adaptive-testing baseline was run; and there are no real candidates and no learning outcomes. The full paper will present these limits alongside the results.